

**GRAPH THEORY:
HOMEWORK 2
BOOK SECTION 1.2.**

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1. Suppose that every vertex of a simple graph G has degree at least 3. Prove that G has a cycle of even length. (Hint: Consider a maximal path $P = v_1v_2 \cdots v_k$)

Proof. Let $P = v_1v_2 \cdots v_k$ be a maximal path in G . By maximality, every neighbor of v_k lies on P ; otherwise, a neighbor outside P could be appended to obtain a longer path. Since $d(v_k) \geq 3$, the vertex v_k has at least two neighbors on P other than v_{k-1} . Choose two of them, say v_i and v_j , with $i < j < k - 1$.

If either $k - i + 1$ or $k - j + 1$ is even, then the corresponding cycle

$$v_i v_{i+1} \cdots v_k v_i \quad \text{or} \quad v_j v_{j+1} \cdots v_k v_j$$

has even length. Otherwise both of these numbers are odd, so i and j have the same parity. In that case

$$v_i v_{i+1} \cdots v_j v_k v_i$$

is a cycle of length $(j - i) + 2$, which is even because $j - i$ is even. Thus G contains a cycle of even length. $\square$

2. Prove that a bipartite graph has a unique bipartition (except for interchanging the two partite sets) if and only if it is connected.

Proof. First suppose that G is connected, and let (A, B) be a bipartition. Fix a vertex $r \in A$. For every vertex x , any r - x path alternates between A and B . Consequently, $x \in A$ exactly when $d(r, x)$ is even, and $x \in B$ exactly when $d(r, x)$ is odd. Thus the choice of the part containing r determines the entire bipartition. The only other choice interchanges A and B , so the bipartition is unique up to interchange.

Conversely, suppose that G is disconnected and let (A, B) be a bipartition. Choose one connected component H of G and interchange the two partite sets only on H ; that is, set

$$A' = (A \setminus V(H)) \cup (B \cap V(H)), \quad B' = (B \setminus V(H)) \cup (A \cap V(H)).$$

There are no edges between H and the other components, so (A', B') is also a bipartition. Since at least one component was not switched, (A', B') is neither (A, B) nor (B, A) . Hence a disconnected bipartite graph does not have a unique bipartition up to interchange. $\square$

3. Let v be a cut-vertex of a simple graph G . Prove that $\overline{G} - v$ is connected.

Proof. Because v is a cut-vertex, $G - v$ has at least two connected components. Let x and y be arbitrary vertices of $G - v$. If they belong to different components of $G - v$, then $xy \notin E(G)$, and hence $xy \in E(\overline{G})$. Thus they are adjacent in $\overline{G} - v$.

If x and y belong to the same component of $G - v$, choose a vertex z in a different component. Neither xz nor yz is an edge of G , so both are edges of $\overline{G}$. Therefore xyz

is a path in $\overline{G} - v$. Every two vertices of $\overline{G} - v$ are thus joined by a path, so $\overline{G} - v$ is connected. $\square$

4. Let G be a simple graph having no isolated vertex and no induced subgraph with exactly two edges. Prove that G is a complete graph.

Proof. We first show that G is connected. If it were disconnected, choose an edge x_1x_2 in one component and an edge y_1y_2 in another. Such edges exist because G has no isolated vertices. The subgraph induced by $\{x_1, x_2, y_1, y_2\}$ would have exactly the two edges x_1x_2 and y_1y_2 , a contradiction.

Now suppose that G is not complete. Choose two nonadjacent vertices u and w . Since G is connected, there is a shortest u - w path

$$u = v_0, v_1, \dots, v_m = w,$$

where $m \geq 2$. By the minimality of this path, v_0v_2 is not an edge. Hence the subgraph induced by $\{v_0, v_1, v_2\}$ has exactly the two edges v_0v_1 and v_1v_2 , again a contradiction. Therefore G must be complete. $\square$

5. Let G be a connected simple graph not having P_4 , or C_3 as an induced sub-graph. Prove that G is a biclique (complete bipartite graph). (Hint: Recall that the distance $d_G(u, v)$ between vertices u and v in G is the shortest path between u and v [assuming that G is connected]. Fix an arbitrary vertex $v \in V(G)$. Partition $V(G)$ using distances to v by letting $L_i = \{u \in V(G) : d_G(v, u) = i\}$ for $i \geq 0$. Why must $L_i = \emptyset$ for $i \geq 3$. Why must G be complete bipartite?)

Proof. Fix a vertex $v \in V(G)$ and, for each $i \geq 0$, let

$$L_i = \{u \in V(G) : d_G(v, u) = i\}.$$

We first claim that $L_i = \emptyset$ for every $i \geq 3$. Indeed, if $u \in L_i$ for some $i \geq 3$, then the first four vertices $v_0 = v, v_1, v_2, v_3$ of a shortest v - u path induce a P_4 . There can be no edge joining two nonconsecutive vertices among them, because such an edge would give a shorter path from v to v_2 or v_3 . This contradicts the hypothesis. Therefore

$$V(G) = L_0 \cup L_1 \cup L_2.$$

The set L_1 is independent: an edge between two of its vertices, together with v , would induce a triangle. The set L_2 is also independent. To see this, suppose $x, y \in L_2$ and xy is an edge. Choose $a \in L_1$ adjacent to x . If ay is an edge, then a, x, y form a triangle. If ay is not an edge, then v, a, x, y induce a P_4 . Both alternatives are impossible.

Finally, every vertex of L_1 is adjacent to every vertex of L_2 . Let $y \in L_2$, and choose $a \in L_1$ adjacent to y . If some $b \in L_1$ were not adjacent to y , then b, v, a, y would induce a P_4 : the only possible additional edges are excluded because L_1 is independent and because $y \notin L_1$. This is impossible.

Let $A = L_1$ and $B = L_0 \cup L_2$. Both A and B are independent, and every vertex of A is adjacent to every vertex of B : vertices of L_1 are adjacent to v , and, as just proved, to every vertex of L_2 . Hence G is the complete bipartite graph with partite sets A and B . $\square$